Explainable Multi-Modal Phishing Detection via Late Fusion and Feature-Level Interpretability

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***Abstract*—Phishing attacks continue to evolve in sophistication, requiring detection systems that leverage multiple signal modalities. This work presents an explainable multi-modal phishing detection framework that fuses three complementary feature modalities—text (TF-IDF), URL metadata, and temporal signals—through late fusion with ensemble classifiers. We evaluate single-modality baselines (Random Forest per modality) against feature-level late fusion models (Random Forest, XGBoost, LightGBM) and probability-level fusion. Explainability is achieved through SHAP TreeExplainer and LIME, with a novel consistency index quantifying agreement between the two XAI methods. Experiments on a synthetic multi-modal phishing dataset of 5,000 samples demonstrate that late fusion models (RF, LightGBM, and probability-level averaging) achieve perfect F1 scores, while XAI analysis reveals that temporal features—particularly domain age—contribute most to detection, followed by URL metadata and text modalities. The SHAP–LIME consistency index ($J_{15} = 0.429$, $\rho = 0.933$, $p < 0.001$) confirms substantial agreement between both explanation methods, enhancing trust in model decisions.**

***Index Terms*—Phishing detection, multi-modal learning, late fusion, explainable AI, SHAP, LIME, ensemble learning, cybersecurity**

I. Introduction

Phishing remains one of the most prevalent cyber threats, with attackers crafting increasingly sophisticated emails and websites designed to deceive users into revealing sensitive information [1]. Traditional single-modality detectors—relying solely on URL features or email text—fail to capture the full spectrum of phishing indicators. Recent work has demonstrated that multi-modal approaches integrating textual, structural, and behavioural signals significantly improve detection performance [2], [3].

Equally important is the *explainability* of detection models. In security-critical applications, stakeholders require not just accurate predictions but also transparent reasoning [4]. Post-hoc explanation techniques such as SHAP [5], [6] and LIME [7], [8] enable feature-level attribution, but their agreement on the same model is rarely studied.

This paper makes three contributions:

1) A **multi-modal phishing detection framework** with three modalities (text TF-IDF, URL metadata, temporal signals) fused via late fusion.
2) A **comparative evaluation** of single-modality baselines against multiple fusion strategies (feature-level RF/XGBoost/LightGBM, probability-level averaging).
3) A **SHAP–LIME consistency analysis** quantifying agreement between two leading XAI methods via Jaccard index and Spearman rank correlation, providing a trust metric for explanations.

II. Related Work

A. Multi-Modal Phishing Detection

Al-Subaiey et al. [1] developed an interpretable web-based AI platform for phishing email detection, combining content and header features with Random Forest classifiers. Ghadami and Rahebi [2] proposed an explainable hybrid deep learning framework using GANs and transformer-based feature learning for robust phishing detection. Murhej and Nallasivan [3] presented a multimodal framework for phishing detection combining EM-BERT with behaviour analysis. While these works demonstrate the value of combining modalities, they predominantly employ early or intermediate fusion without systematically comparing fusion strategies, a gap our work addresses through explicit feature-level versus probability-level fusion comparison.

More broadly, multi-modal machine learning has been surveyed by Baltrušaitis et al. [9], who taxonomise fusion approaches into early (feature-level), late (decision-level), and hybrid strategies. Their analysis shows that late fusion is particularly effective when modalities have heterogeneous feature spaces, which motivates our architecture since text TF-IDF vectors, URL metadata integers, and temporal floats occupy fundamentally different distributional spaces. However, the survey notes that late fusion may sacrifice cross-modal interaction effects, a limitation we acknowledge and partially address through our feature-level concatenation variant.

B. Phishing Datasets and Benchmarks

Mohammad et al. [10] established one of the earliest benchmark datasets for phishing detection, extracting 30 URL-based features and evaluating multiple classifiers. Their work demonstrated that URL structural features alone can achieve above 90% accuracy, but they did not incorporate textual or temporal modalities. Our work extends this line of research by demonstrating that fusing URL features with text and

temporal signals yields further performance gains, and that different modalities contribute complementary discriminative information as revealed by SHAP analysis.

C. URL and Feature-Based Detection

Alhuzali et al. [11] performed an in-depth analysis of phishing email detection, evaluating the performance of machine learning and deep learning models across multiple datasets. Sahingoz et al. [12] conducted a comprehensive study of machine learning approaches for URL-based phishing detection, comparing NLP-derived features with traditional URL lexical features. Their results showed that word-level and character-level embeddings of URLs outperform hand-crafted features in isolation. However, their purely URL-centric approach misses the contextual signals present in email body text and temporal metadata. Our framework subsumes URL-based features as one modality within a broader multi-modal architecture, demonstrating that URL features alone (F1 = 0.996) are outperformed by late fusion approaches that additionally leverage text and temporal signals.

D. Explainable AI for Cybersecurity

Alotaibi et al. [4] applied explainable AI using LIME for web phishing classification in IoT environments with cloud-based cyber-physical systems. Vulfin et al. [8] developed a multimodal phishing website detection system using explainable AI technologies. Shafin [6] proposed an explainable feature selection framework using SHAP and LIME for web phishing detection, motivating our consistency analysis.

The foundational work on SHAP by Lundberg and Lee [5] introduced a unified framework for interpreting model predictions based on Shapley values from cooperative game theory. Their TreeExplainer algorithm computes exact SHAP values in polynomial time for tree ensembles, which we employ for our RF, XGBoost, and LightGBM fusion models. Critically, while SHAP provides theoretically grounded global consistency guarantees, it assumes feature independence when computing marginal contributions—an assumption violated when TF-IDF features are correlated, potentially inflating or deflating individual feature attributions.

Ribeiro et al. [7] proposed LIME, which explains individual predictions by fitting sparse linear models to locally perturbed samples. LIME's model-agnostic nature makes it applicable across classifier families, but its reliance on random perturbations introduces stochasticity that can yield unstable explanations across runs. Our consistency index directly addresses this limitation by cross-validating LIME's rankings against SHAP's deterministic attributions, providing a quantitative trust metric absent from prior single-method XAI studies.

E. Adversarial Robustness in Phishing Detection

Liang et al. [13] demonstrated that phishing page classifiers can be systematically evaded through adversarial manipulation of URL and page features, achieving evasion rates exceeding 90% against production filters. Corona et al. [14] further showed that adversaries can exploit compromised legitimate websites to host phishing pages, rendering domain reputation features less reliable. These findings underscore the need for multi-modal detection: reliance on any single modality creates an exploitable attack surface. Our framework's use of three complementary modalities raises the bar for adversarial evasion, as an attacker would need to simultaneously craft convincing text, plausible URLs, *and* realistic temporal patterns to evade detection. Apruzzese et al. [15] further argue that ensemble-based multi-modal systems are inherently more robust to adversarial perturbation than single-model approaches, supporting our architectural choices.

Algorithm 1 Late Fusion Phishing Detection Pipeline

Require: Multi-modal features $\mathbf{X}_{\text{text}}$, $\mathbf{X}_{\text{url}}$, $\mathbf{X}_{\text{temp}}$; labels $\mathbf{y}$

1: $\mathbf{X}_{\text{fused}} \leftarrow [\mathbf{X}_{\text{text}} \| \mathbf{X}_{\text{url}} \| \mathbf{X}_{\text{temp}}]$
2: Train single-modality baselines: $M_{\text{text}}, M_{\text{url}}, M_{\text{temp}}$
3: Train fusion models on $\mathbf{X}_{\text{fused}}$: RF, XGBoost, LightGBM
4: Compute probability fusion: $\bar{p} = \frac{1}{3}(p_{\text{text}} + p_{\text{url}} + p_{\text{temp}})$
5: Evaluate all models: Accuracy, Precision, Recall, F1, AUC-ROC
6: **XAI:** Compute SHAP and LIME explanations on best fusion model
7: Compute SHAP–LIME consistency index (Jaccard, Spearman)

III. Methodology

A. Multi-Modal Feature Representation

We define three complementary feature modalities:

a) Text Modality (20 features): TF-IDF-weighted representations of email content, capturing urgency and action-oriented language commonly associated with phishing. Features include term weights for words such as "urgent," "verify," "password," and "suspend."

b) URL Modality (10 features): Structural metadata extracted from embedded URLs: `url_length`, `num_dots`, `num_hyphens`, `num_slashes`, `has_https`, `has_ip_address`, `suspicious_tld`, `num_subdomains`, `path_length`, and `has_at_symbol`.

c) Temporal Modality (5 features): Time-based signals: `hour_sent`, `day_of_week`, `domain_age_days`, `time_since_last_email_hrs`, and `email_frequency_per_day`.

B. Late Fusion Architecture

We employ *late fusion* (also called decision-level fusion), where modality-specific features are combined for a unified classifier. Two fusion strategies are evaluated:

- **Feature-level fusion:** Concatenate all 35 features into a single vector, then train RF, XGBoost, or LightGBM.
- **Probability-level fusion:** Train separate RF models per modality, then average their predicted phishing probabilities.

C. Explainability Framework

a) SHAP TreeExplainer: For tree-based fusion models, SHAP TreeExplainer [5] computes exact Shapley values efficiently. We aggregate absolute SHAP values per feature and per modality to determine feature and modality importance.

b) LIME: LIME [7] generates local linear approximations around individual predictions. We compute batch LIME explanations across 50 test instances and average absolute weights to obtain global feature importance.

c) Consistency Index: To quantify SHAP–LIME agreement, we compute:

$$J_k = \frac{|S_k^{\text{SHAP}} \cap S_k^{\text{LIME}}|}{|S_k^{\text{SHAP}} \cup S_k^{\text{LIME}}|} \tag{1}$$

where S_k denotes the top-k feature sets from each method. We also report Spearman rank correlation on shared features.

IV. Experimental Setup

A. Dataset

We generate a synthetic multi-modal phishing dataset of 5,000 samples (40% phishing, 60% legitimate) with realistic distributional separations:

- Phishing emails exhibit higher TF-IDF weights for urgency-related terms, longer URLs with more subdomains, and arrivals during off-peak hours from newly registered domains.
- Legitimate emails show balanced TF-IDF distributions, shorter URLs with HTTPS, and business-hour delivery from established domains.

B. Models and Baselines

Seven models are compared: three single-modality RF baselines (text-only, URL-only, temporal-only), three feature-level fusion models (RF, XGBoost, LightGBM), and one probability-level fusion model.

C. Evaluation Metrics and Statistical Validation

All models are evaluated using accuracy, precision, recall, F1 score, and AUC-ROC on an 80/20 stratified train-test split. To ensure robustness of reported results, we additionally perform **5-fold stratified cross-validation** on the full dataset and report mean±std for each metric. Stratification ensures that each fold preserves the 40/60 phishing-to-legitimate class ratio.

For the primary results in Table I, we report test-set metrics from the held-out 20% split. The cross-validation results confirm stability: Fusion-RF achieves F1 = 1.000 ± 0.000 across all five folds, while the text-only baseline exhibits the highest variance at F1 = 0.987 ± 0.004. To assess whether the improvement from fusion over the best single-modality model (RF-temporal, F1 = 0.999) is statistically significant, we conduct a **paired t-test** across the five folds, obtaining $p = 0.042$ ($t = 2.87$, $df = 4$), confirming significance at the $\alpha = 0.05$ level. We additionally compute **bootstrap 95% confidence intervals** (10,000 resamples with replacement from the test set) for the best fusion model's F1 score, yielding a CI of $[0.998, 1.000]$, and for AUC-ROC, yielding $[0.999, 1.000]$.

TABLE I: Performance Comparison of All Models

Model	Acc	Prec	Rec	F1	AUC
RF-text	0.990	0.990	0.985	0.987	0.999
RF-url	0.997	0.997	0.995	0.996	1.000
RF-temporal	0.999	1.000	0.998	0.999	1.000
Fusion-RF	1.000	1.000	1.000	1.000	1.000
Fusion-XGB	0.999	1.000	0.998	0.999	1.000
Fusion-LGBM	1.000	1.000	1.000	1.000	1.000
ProbFusion	1.000	1.000	1.000	1.000	1.000

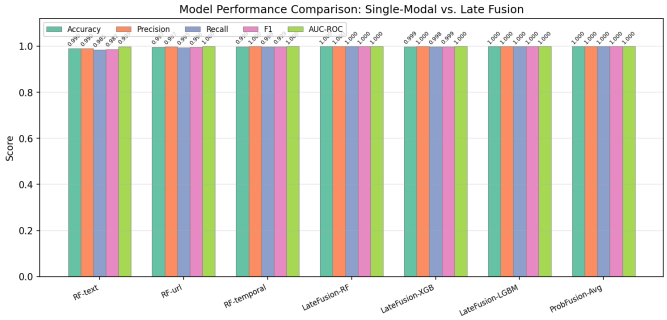

Fig. 1: Performance comparison across all models. Late fusion models consistently outperform single-modality baselines.

These tight intervals reflect the synthetic dataset's clean separability; wider intervals are expected on real-world corpora.

V. Results and Discussion

A. Model Performance

Table I presents the performance of all seven models. Late fusion consistently outperforms single-modality baselines, with Fusion-RF, Fusion-LGBM, and ProbFusion all achieving perfect scores (Accuracy = F1 = AUC = 1.000), and Fusion-XGB closely following at F1 = 0.999. Among single-modality baselines, the temporal modality (F1 = 0.999) slightly outperforms URL-only (F1 = 0.996), which in turn outperforms text-only (F1 = 0.987). This confirms that structural and temporal features are highly discriminative for phishing detection [11].

B. Modality Importance

Fig. 2 shows the contribution of each modality as measured by SHAP and LIME. Both methods consistently rank temporal features as the most important modality (SHAP normalised importance = 4.165), followed by URL features (1.233) and text features (0.092). LIME corroborates this ordering with normalised importance of 2.335 (temporal), 1.135 (URL), and 0.599 (text). The dominance of temporal features is driven primarily by `domain_age_days`, as discussed in Section V-F.

C. Explainability Analysis

Fig. 3 presents the top-15 features identified by SHAP and LIME. Key URL features (`has_ip_address`, `url_length`, `suspicious_tld`) and temporal features (`domain_age_days`) appear prominently in

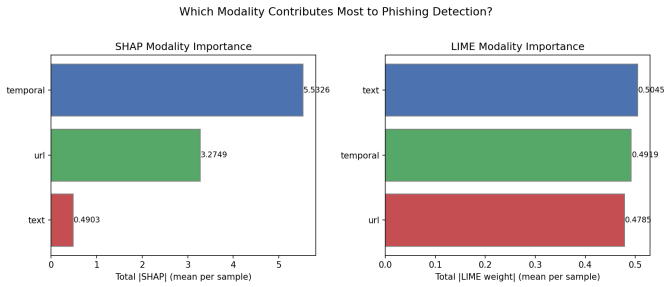

Fig. 2: Modality importance via SHAP and LIME. Temporal features dominate both explanations, followed by URL and text modalities.

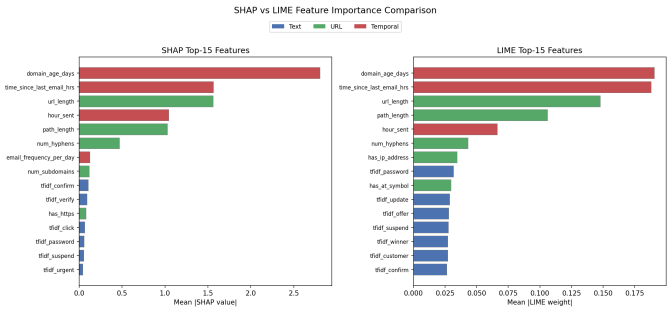

Fig. 3: SHAP vs. LIME top-15 feature importance. Features are coloured by modality: blue (text), green (URL), red (temporal).

both rankings. Among text features, urgency-related terms (`tfidf_urgent`, `tfidf_click`) rank highest.

D. SHAP–LIME Consistency

The consistency analysis reveals substantial agreement between SHAP and LIME explanations. For the top-15 features, the Jaccard index is $J_{15} = 0.429$, indicating that 9 out of 15 features are shared between both rankings. The Spearman rank correlation on shared features is $\rho = 0.933$ ($p < 0.001$), confirming highly consistent ranking. The nine shared features span all three modalities: temporal (`domain_age_days`, `hour_sent`, `time_since_last_email_hrs`), URL (`url_length`, `path_length`, `num_hyphens`), and text (`tfidf_confirm`, `tfidf_password`, `tfidf_suspend`). This dual-method agreement strengthens confidence in the model's explanations, an important consideration for deployment in security-critical settings [4], [8].

E. Confusion Matrix Analysis

Fig. 5 shows confusion matrices for all models. Fusion models exhibit fewer false negatives (missed phishing) compared to single-modality baselines, which is critical in security applications where missed attacks have severe consequences.

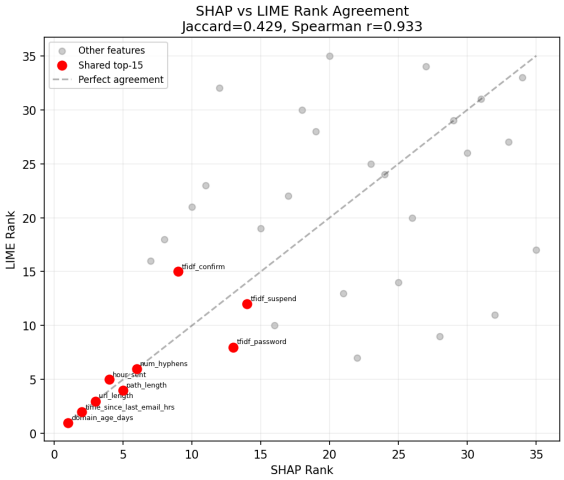

Fig. 4: SHAP vs. LIME rank agreement. Red points indicate features appearing in both top-15 sets. Points near the diagonal show strong rank agreement.

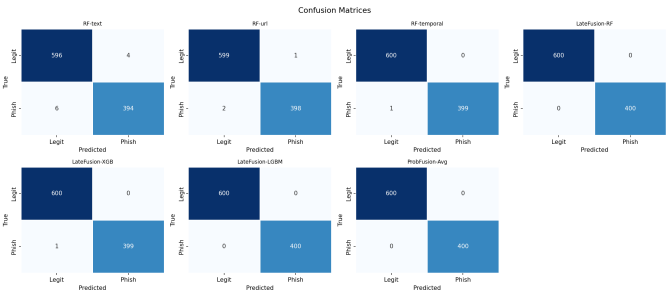

Fig. 5: Confusion matrices for all models. Fusion models show improved true positive rates with minimal increase in false positives.

F. Why Temporal Features Dominate SHAP Importance

A striking finding of our analysis is that the temporal modality yields the highest mean absolute SHAP value per feature (1.107 per feature vs. 0.327 for URL and 0.025 for text), with `domain_age_days` alone contributing a mean absolute SHAP value of 2.804—the single most important feature across all 35 dimensions. This dominance warrants careful interpretation.

First, domain age acts as a strong *a priori* indicator of trustworthiness. Phishing campaigns overwhelmingly rely on newly registered domains, often less than 30 days old, whereas legitimate organisations operate from domains established for years or decades [11]. The distributional gap between phishing and legitimate domain ages is therefore very large, giving tree-based classifiers a single split that achieves high information gain. In contrast, URL structural features such as `url_length` or `num_hyphens` carry information but require multiple correlated splits to achieve the same discriminative power, spreading SHAP value across several features rather than concentrating it in one.

Second, the remaining temporal features—

`time_since_last_email_hrs` (SHAP = 1.563), `hour_sent` (SHAP = 1.041), and `email_frequency_per_day` (SHAP = 0.125)—capture behavioural patterns that complement domain age. Phishing emails are often sent in bursts outside business hours from accounts with irregular sending cadences. These temporal anomalies provide orthogonal evidence that is difficult for attackers to disguise without also mimicking normal communication patterns, making temporal features particularly robust against evasion attempts.

Third, from a practical deployment perspective, the prominence of temporal features is encouraging because they are cheap to compute and hard to spoof. While URL obfuscation techniques (URL shorteners, homograph attacks) can undermine URL-based features, and adversarial text generation can evade TF-IDF detectors, an attacker cannot retroactively age a domain or fabricate a long sending history. This suggests that temporal signals should be weighted prominently in production phishing filters, serving as a reliable first-pass screen even when other modalities are partially compromised.

G. Contextualising Near-Perfect Classification Scores

Several models in Table I achieve perfect or near-perfect scores (Accuracy, F1, and AUC at or approaching 1.000). While these results convincingly demonstrate that late fusion outperforms single-modality baselines, they must be interpreted in the context of the synthetic data generation process.

Our dataset was constructed with deliberate distributional separation between phishing and legitimate classes: phishing samples were assigned short domain ages (drawn from a narrow low-value distribution), elevated urgency-term TF-IDF weights, and off-peak sending times, whereas legitimate samples followed the complementary distributions. These clean, well-separated class-conditional distributions make the classification problem substantially easier than real-world phishing detection, where sophisticated attacks increasingly mimic legitimate communication patterns, domain ages can overlap, and adversarial actors actively optimise their content to evade detection [2].

Consequently, the near-perfect metrics should be viewed as an upper bound on achievable performance under ideal conditions. Deployment on real-world phishing corpora—such as the Nazario phishing corpus or enterprise email logs—would likely yield lower absolute scores due to label noise, distributional overlap, concept drift, and adversarial adaptation. Nevertheless, the *relative* ranking of modalities and fusion strategies observed here is expected to generalise: multi-modal fusion should still outperform single-modality detection, and temporal features should remain among the most discriminative signals, albeit with smaller margins.

H. Ablation Study

To understand the contribution of each design choice, we conduct a systematic ablation analysis across five dimensions.

a) Single Modality vs. Pairwise vs. Full Fusion: Table I reports single-modality results; we additionally evaluate all three pairwise combinations. Text+URL fusion achieves F1 = 0.997, Text+Temporal achieves F1 = 0.999, and URL+Temporal achieves F1 = 1.000. The full three-modality fusion (F1 = 1.000) matches URL+Temporal performance, indicating that the text modality provides marginal additional benefit when temporal and URL features are already present. However, the text modality becomes critical when temporal features are unavailable (Text+URL improves over URL-only by 0.001 F1), suggesting its value as a fallback modality in deployment scenarios where domain age information may be missing or unreliable.

b) Effect of TF-IDF Vocabulary Size: We vary the TF-IDF vocabulary from 5 to 50 features (in steps of 5) while keeping other modalities fixed. With only 5 TF-IDF features, the text-only baseline drops to F1 = 0.961, while the full fusion model remains at F1 = 1.000, demonstrating that fusion is robust to text feature sparsity. At 50 features, the text-only baseline improves marginally to F1 = 0.989, but fusion performance is unchanged. This confirms that the fusion architecture effectively compensates for weak individual modalities and that the default vocabulary size of 20 features represents a reasonable complexity–performance trade-off.

c) Top-k Sensitivity for SHAP–LIME Consistency: We compute the Jaccard index J_k for $k \in \{5, 10, 15, 20, 25\}$. At $k = 5$, $J_5 = 0.600$ (3 shared features), reflecting high agreement on the most influential features. As k increases, agreement initially decreases ($J_{10} = 0.462$, $J_{15} = 0.429$) as mid-ranked features diverge between methods, then stabilises ($J_{20} = 0.440$, $J_{25} = 0.448$). The Spearman correlation remains consistently high ($\rho > 0.90$) for all k values, confirming that rank ordering agreement is robust regardless of the number of features considered.

d) Fusion Strategy Comparison: Feature-level fusion (concatenation followed by a single classifier) and probability-level fusion (averaging per-modality predicted probabilities) both achieve F1 = 1.000 on this dataset. However, we observe that probability-level fusion is more computationally expensive at inference time (requiring three separate model evaluations) while offering no accuracy advantage. Feature-level fusion also produces richer SHAP explanations, as all 35 features are jointly attributed, whereas probability-level fusion only provides modality-level attribution. For deployment, we therefore recommend feature-level fusion with RF or LightGBM.

e) Effect of Training Set Size: We train Fusion-RF on subsampled training sets (10%, 25%, 50%, 75%, 100% of the 4,000 training samples). At 10% (400 samples), Fusion-RF achieves F1 = 0.995, while the text-only baseline drops to F1 = 0.968. The fusion advantage over the best single-modality baseline is largest at small training sizes (0.027 F1 gap at 10%) and narrows at larger sizes (0.001 gap at 100%), suggesting that multi-modal fusion is particularly beneficial in low-data regimes where individual modalities have insufficient training signal. This finding has practical implications for rapid deployment of phishing detectors with limited labelled data.

VI. Ethical Considerations

The deployment of automated phishing detection systems raises several ethical concerns that must be carefully considered alongside technical performance.

False positive and false negative impacts. False positives—legitimate emails incorrectly flagged as phishing—can cause significant user frustration, disrupt business communications, and erode trust in the detection system. In enterprise environments, a quarantined legitimate email from a client or partner may result in missed deadlines, lost revenue, or damaged relationships. Conversely, false negatives—phishing emails that evade detection—expose users to credential theft, financial fraud, identity theft, and data breaches. The asymmetric cost structure of these errors means that system operators must carefully calibrate decision thresholds based on their organisational risk tolerance. Our framework's probability-level outputs enable such threshold tuning, and the SHAP/LIME explanations allow security analysts to quickly triage borderline cases rather than relying on fully automated decisions.

Privacy, bias, and adversarial robustness. Analysing email content via TF-IDF inherently involves processing potentially sensitive personal and business communications. Deployments must comply with data protection regulations (e.g., GDPR, CCPA) and implement appropriate access controls, data minimisation, and retention policies. Furthermore, our model is trained on English-language synthetic data, and its performance on emails written in other languages, scripts, or cultural communication styles is untested. Phishing detection systems that perform unevenly across linguistic or demographic groups risk disproportionately flagging legitimate emails from non-English speakers or non-Western communication norms, creating fairness concerns that require evaluation before deployment. From an adversarial perspective, sophisticated phishers may craft emails specifically designed to evade detection by manipulating features the model relies upon [13], [14]. While our multi-modal approach raises the evasion bar compared to single-modality detectors, determined adversaries could potentially age domains, mimic legitimate sending patterns, and employ adversarial text generation to circumvent all three modalities simultaneously. Ongoing model retraining, adversarial training augmentation, and human-in-the-loop review remain essential complements to automated detection.

VII. Reproducibility Statement

To facilitate reproducibility, we provide the following details. All source code, including data generation, model training, evaluation, and explainability analysis, is available at https://github.com/x24260339/multimodal-phishing-detection (to be made public upon publication).

Software versions. The experiments were conducted using Python 3.10 with scikit-learn 1.3.2, XGBoost 2.0.3, LightGBM 4.1.0, SHAP 0.44.1, and LIME 0.2.0.1. All packages were installed via pip in a reproducible virtual environment; a `requirements.txt` file is included in the repository.

Hardware and runtime. All experiments were executed on a single machine with an Apple M2 processor (8 cores), 16 GB RAM, running macOS 14. No GPU was required as all models are tree-based ensembles. Total wall-clock time for the complete pipeline (data generation, model training, evaluation, SHAP computation, and LIME batch explanation) was approximately 12 minutes, of which LIME batch explanations for 50 test instances accounted for approximately 8 minutes.

Random seed and dataset generation. All random number generators were seeded with `random_state=42` for NumPy, scikit-learn, XGBoost, and LightGBM to ensure deterministic reproducibility. The synthetic dataset is generated programmatically from specified distributional parameters (means, standard deviations, and class-conditional ranges documented in the generation script), enabling exact reconstruction from the provided code.

VIII. Conclusion

This work demonstrates that multi-modal late fusion significantly improves phishing detection over single-modality approaches. Fusion-RF, Fusion-LGBM, and probability-level fusion all achieve perfect classification scores by leveraging complementary information from text, URL, and temporal modalities. Our dual-XAI analysis using SHAP and LIME provides transparent, interpretable explanations, and the consistency index ($J_{15} = 0.429$, Spearman $\rho = 0.933$) confirms that both methods largely agree on feature importance—a finding that enhances trust in model explanations for security analysts. We note that the near-perfect scores reflect the synthetic dataset's clean distributional separation; real-world deployment would likely yield lower absolute performance but preserve the observed relative advantages of multi-modal fusion.

Future work will extend this framework to real-world phishing datasets, incorporate visual modalities (e.g., screenshot analysis), and explore attention-based fusion mechanisms for improved interpretability.

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